

Placement techniques for composite in CI I

✓ Incremental technique

It is including place (and light cure) the composite incrementally to ensure maximum polymerization and possibly to reduce the negative effects of polymerization shrinkage.

A typical Class I tooth preparation will have a high C-factor of 5/1 (i.e., five bonded surfaces—pulpal, facial, lingual, mesial, and distal—versus one unbonded surface—occlusal). the higher the C-factor of a tooth preparation, the higher the potential for composite polymerization shrinkage stress, as the composite shrinkage deformation is restricted by the bonded surfaces. the interface most likely to be adversely affected by high C-factor stresses is at the pulpal wall as the anatomy of dentin in this area makes establishment of a durable interface more difficultly.

Incremental insertion and light curing of the composite may reduce the negative C-factor effects for Class I composite restorations.

Clinical steps:

Posterior composites should be placed incrementally to facilitate proper light curing and development of correct anatomy, especially in Class I direct composite restorations, the anatomic references of the occlusal unprepared tooth structure should guide the placement and shaping of the composite increments.

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very deep portions of the tooth preparation are restored first, with increments of no more than 2 mm in thickness (Fig. 1). the “enamel layer” of the restoration, that is, the occlusal increment(s), should be placed using an anatomic layering technique.



Fig 1: Incremental technique for application of composite in C I I

Typically, the operator places and light cures one increment per cusp at a time and continues to place subsequent increments until the preparation is filled and the occlusal anatomy is fully developed the uncured composite can be shaped against the unprepared cusp inclines as visual guides (see fig 2 C-F), which will result in a very natural anatomic contour. Sculpture of the occlusal elements of the restoration following the anatomic references of the tooth respects the tooth's anatomy and occlusion and minimizes the need for contouring and finishing after the composite is polymerized.

Furthermore, this technique prevents potential damage to the enamel adjacent to the restoration margins because it minimizes the need to use rotary instruments to

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remove excess composite extending beyond those margins. Any suitable composite hand instruments can be used with the anatomic layering technique.

Fine composite spatulas and the tine of an explorer can be used to further develop or refine the anatomy of uncured composite increments, microbrushes also can be used to smooth uncured composite against the preparation margin, but these should never be saturated with adhesives. Once the composite is fully cured, if additional contouring is needed, the restoration can be finished immediately after the last increment is cured using appropriate finishing instruments.

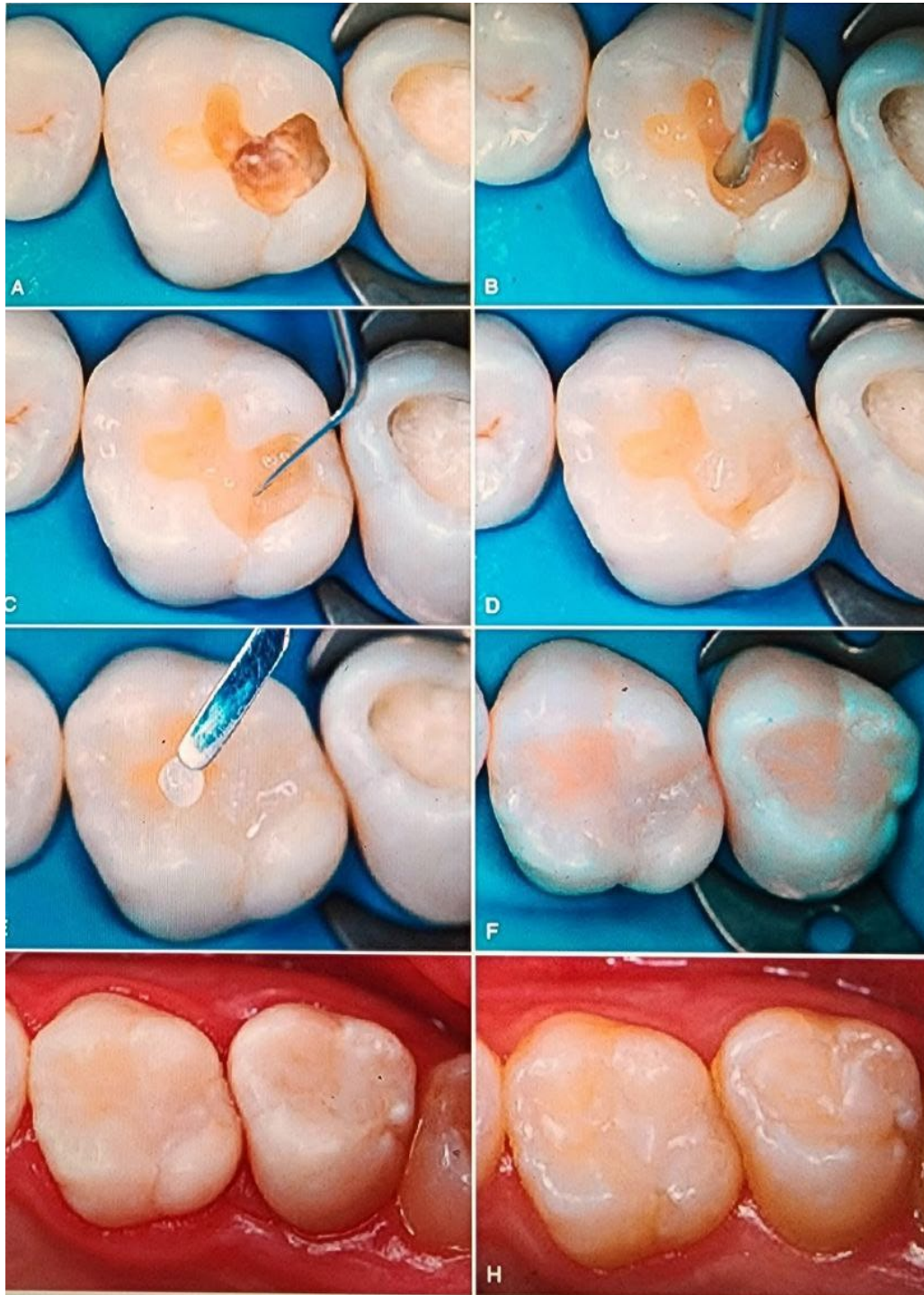
✓ Sandwich technique

The use of an RMGI (Resin Modified Glass Ionomer) liner or a flowable composite liner were thought to reduce the effects of polymerization shrinkage stress because of their favorable elastic modulus (a more elastic material would more effectively absorb polymerization stresses) (see fig 2 B),

The potential advantages of this technique are

- (1) the RMGI material bonds to the dentin without the need for a dental adhesive;
- (2) the RMGI material potential for fluoride release (potential anti-cariogenicity), may provide a better seal when used in cases where the preparation extends gingivally onto root structure; and
- (3) the favorable elastic modulus of the RMGI may reduce the effects of polymerization shrinkage stresses.

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• **Fig. 8.10** Class I composite incremental insertion. A, Tooth preparation for Class I direct composite restoration. B, After a resin-modified glass ionomer base is placed, the first composite increment is inserted and light cured. C-F, Composite is inserted and light cured incrementally, using cusp inclines as anatomic references to sculpt the composite before light curing. G, Completed restorations. H, At 5-year follow-up.

Fig 2: C I I composite insertion techniques

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✓ Bulk-fill technique

“Bulk-fill composites” have gained increased interest in recent years. these materials are available as *flowable base bulk-fill composites* and *full-body bulk-fill composites*.

Clinical steps:

Flowable base bulk-ill composites require a conventional composite as the occlusal increment and therefore are used for dentin replacement only,

whereas full-body bulk-fill composites can replace dentin and enamel in a single increment. their use can expedite the restorative procedure as increments of up to 4 mm in thickness are often suggested.

Note: A matrix is usually not necessary for Class I direct composite restorations, even when facial and lingual surface grooves are included, Composite insertion hand instruments used to place the composite material incrementally

Finishing and Polishing of the Composite

If the composite is carefully placed and shaped before light curing, as described, additional finishing with burs is substantially minimized. However, in many cases, refined finishing may be needed, especially when occlusion adjustments are necessary.

The occlusal surface is shaped with a round or oval carbide finishing bur or similarly shaped finishing diamond. Polishing is accomplished with appropriate polishing cups, points, or both after the occlusion is adjusted as necessary (Fig. 3).

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• **Fig. 8.12, cont'd** E, Incremental placement of composite. F, Rubber dam is removed and occlusion checked. G, Buccal view, a finishing fluted bur is used to selectively adjust the occlusion. H, Polishing with brush and diamond paste. I, Completed restoration.

Fig 3: Composite finishing and polishing

Placement techniques for composite in Cl II

Retainers & matrix band application

One of the most important steps in restoring Class II preparations with direct composites is proper matrix selection and setup.

A matrix band is necessary to assist in the development of the appropriate axial tooth contours, in Class II composites are almost totally dependent on the contour and position of the matrix for establishing appropriate proximal contacts, Generally, the matrix band is applied before adhesive placement. An Ultra-thin metal matrix band generally is preferred for the restoration of a Class II composite because it is thinner than a typical metal band and can be contoured better than a clear polyester matrix.

Types of retainers & matrix band:

Ivory No.8 and ivory No.9 (Tofflemire retainer): these retainers can be used in MOD cavities, the band surrounds the tooth and the retainer lie in mucobuccal fold

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Fig 4: Tofflemire retainer

Ivory no.1: used to surround tooth from one surface MESIAL or DISTAL fig 5.



Fig 5: Ivory no. 1

Sectional metallic matrices: are relatively easy to use, very thin, and come in different sizes so as to accommodate for varied occluso-gingival heights of the proximal box fig 6.

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Fig 6: Sectional metallic matrices

Note: Regardless of the type of matrix system used, the matrix material should extend at least 1 mm beyond the gingival margin (gingivally) and 1mm above marginal ridge of the restoration (occlusally).

Wedge:

A wedge is needed at the gingival margin to

(1) hold the matrix in position, (2) provide slight separation of the teeth, and (3) prevent a gingival overhang of the composite material.

Several types of commercial wedges are available in assorted sizes. An anatomical (triangle-shaped) wedge of compatible size is indicated for most preparations. the wedge is kept as short as possible so as not to limit operator access to the preparation during insertion of the restorative material, and should also not push the matrix onto the preparation space fig 7.

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For posterior restorations, the wedge is placed from the larger to the smaller interproximal gingival embrasure, typically from a *lingual* approach, just apical to the gingival margin. the wedge should engage (stabilize) the matrix band.



Fig7: Wedge

Placement of composite in CI II:

Many techniques have been described for the restoration of the CI II, research comparing different insertion and light-curing techniques is not conclusive and no single technique has been universally accepted.

Note: It is best to restore the proximal box portion of the preparation first.

✓ An oblique Incremental technique

It is important to place and light cure the composite incrementally to maximize the curing potential and to reduce the negative effects of polymerization shrinkage.

Clinical steps:

The number of increments will depend on the size of the proximal box, hand instruments or a “composite gun” may be used to insert the composite material, the first increment should be placed along the gingival floor and should extend slightly

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up the facial (or lingual) wall (see Fig. 8). this increment (or increments for a large box) should be no more than 2 mm thick because it is the farthest increment from the curing light and the most critical in establishing a proper gingival seal.

A second increment is then placed against the lingual (or facial) wall to restore about two thirds of the box. the final increment is then placed to complete the proximal box and develop the marginal ridge. Subsequent additions, if needed, are made and light cured (usually not exceeding 2 mm in thickness at a time) until the proximal box is fully restored (see Fig. 8). Increments should be light cured Regardless of the number of increments needed, when restoring the proximal box, an effort should be made to develop the anatomy of marginal ridge without excessive composite, to reduce the amount of rotary instrumentation required during finishing.

When the proximal box is completed, the occlusal step of the preparation is restored exactly as it was described for the Class I direct composite restoration—that is, using an anatomic layering technique.

Appropriate proximal contact intensity can also be better achieved when composite is applied in increments. the matrix can be held in physical contact with the adjacent proximal surface while the contact-related increment of composite is light cured. A hand instrument with a large surface area (e.g., a small football-shaped or round burnisher) is well suited for that purpose. Once this increment is cured, the proximal contact is established, and remaining increments can be inserted and light cured. the matrix is removed and the restoration is light cured from the facial and lingual directions. the restoration is finished and polished immediately after the last increment is cured.

Composite resin placement may be made more difficult by the stiffness and stickiness of some composite materials. Pre-heated composite up to 68°C (155°F)

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by special composite warmers prior to insertion in the preparation may help overcome these problems.

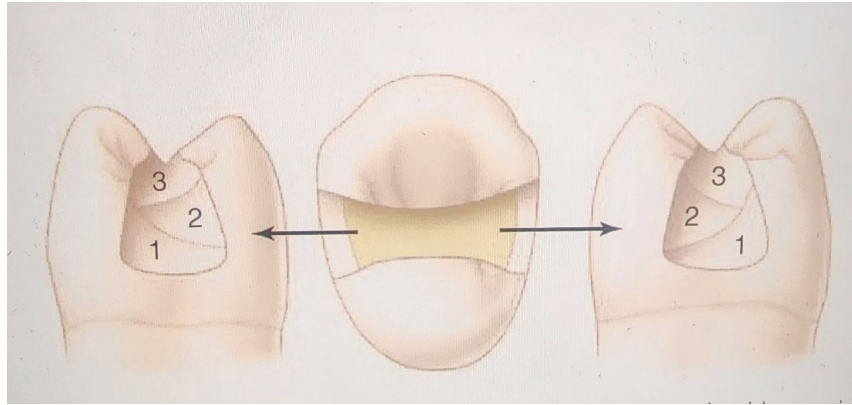


Fig 8: Oblique incremental technique in CI II composite

✓ Sandwich technique

When a stiffer or “packable” high-viscosity composite is used for the restoration of the proximal box, an increment of a flowable composite may be first placed in close proximity to the external margins of the proximal box so as to improve marginal adaptation of the restoration.

✓ Bulkfill

Self-cured and dual-cured composite resin materials are frequently used for large *composite foundations* because these can be injected in the preparation in a single increment. However, it is recommended that even when dual-cured composites are used, they be carefully light cured during and after the final placement.

Finishing and Polishing of the Composite

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Finishing can be initiated immediately after the composite material has been fully light cured (Fig. 9). If the occlusal anatomy was developed the need for additional contouring is greatly minimized. If finishing is needed, the occlusal surface is shaped with a round or oval, 12-bladed carbide finishing bur or finishing diamond. Excess composite is removed at the proximal margins and embrasures with a lame-shaped, 12-bladed carbide finishing bur or finishing diamond and abrasive discs (see Fig. 9 B).

Any overhangs at the gingival area are removed with a No. 12 surgical blade mounted in a Bard-Parker handle with light shaving strokes to remove the excess. Narrow finishing strips may be used to smooth the gingival proximal surface. Care must be exercised in maintaining the position of the finishing strips gingival to the proximal contact area to avoid inadvertent removal of the composite contact with the adjacent proximal surface. the rubber dam (or other means of isolation) is removed, and the occlusion is evaluated for proper contact.

Further adjustments are made, if needed, and the restorations are polished with appropriate polishing points, cups, brushes, or discs.

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Fig 9: Finishing and polishing for composite restoration.